

Sayantan Bhattacharya

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EDUCATION

Techno Main Salt Lake <i>B.Tech in Computer Science and Engineering (AI and Machine Learning)</i>	Kolkata, India CGPA: 7.13 / 10 Expected June 2027
Relevant coursework: Machine Learning, Deep Learning, NLP, Neural Networks, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Database Management Systems, Web Development, Computer Networks, Operating Systems	
Test Scores: GRE: 304/340 - 156Q,148V,4.0 A.W.A TOEFL: 102/120	August 27, 2026 September 03 2026

EXPERIENCE

Independent Technical Contractor <i>Self-Employed — Video Editing, India</i>	2022 — 2026
- Automated a multi-stage post-production pipeline using scripted DaVinci Resolve workflows, reducing per project processing time by 40% across 50+ client deliverables — applied systematic optimization and iterative testing to identify and eliminate bottlenecks in a complex, multi-input system. - Managed end-to-end delivery of 30+ long-form projects across international clients by building a structured scoping and revision framework, achieving zero late deliveries over 5 years. - Designed reusable modular templates for color grading, motion, and audio processing by abstracting recurring patterns into parameterized workflows, scaling output 3x without proportional time increase.	

PROJECTS

PassME — Full-Stack Password Manager <i>Python, Flask, SQLite, HTML, CSS, JavaScript</i>	2025 CS50x Final Project
- Achieved zero critical vulnerabilities across all OWASP Top 10 attack vectors by implementing bcrypt hashing, Fernet encryption, and parameterized queries throughout the authentication and storage pipeline. - Reduced credential retrieval time to under 100ms by designing an indexed SQLite schema with optimized CRUD operations, supporting password generation, copy-to-clipboard, and session management. - Delivered a fully functional, security-compliant password manager as a solo developer, verified through manual penetration testing of all input surfaces. Youtube Demo : youtube.com/watch?v=X2SUDquMxqc	
Semantic Search RAG Chatbot — Two-Stage Retrieval Experiment <i>Python, Flask, ChromaDB, LangChain, Groq (LLaMA-3.3 70B), Scrapy, HuggingFace Transformers</i>	2026 Personal Project
- Improved retrieval precision over single-stage cosine similarity search by designing a two-stage pipeline — ChromaDB vector search over 50 candidates followed by CrossEncoder reranking (ms-marco-MiniLM-L-6-v2) — with a Scrapy and Playwright crawler using SHA-256 hashing to avoid redundant re-indexing. - Enabled coherent multi-turn interaction via a conversation memory module preserving the last 3 Q&A pairs, generating cited answers with LLaMA-3.3 70B via Groq API across PDF, URL, and crawled web sources. GitHub: github.com/PopcornKitsilano/rag-chatbot	
BasketBawl — Basketball Hooping Game <i>MIT Scratch</i>	2024 CS50x PSET0
- Built a real-time score-tracking game with keyboard-driven player movement (A/D) and shooting (1), 2 points per successful shot. - Implemented conditional game flow with name-based prompts and graceful exit handling for non-affirmative user responses. Link: scratch.mit.edu/projects/1061387522	

Razer Crimson Royale — Custom Hardware Lighting Engine

Python, Razer Chroma REST API

2026
Personal Project

- Built a real-time per-key animation engine delivering 20 effects at 7fps by implementing HSV color math, directional transforms, and a session-based REST protocol — demonstrating applied systems programming outside coursework.
- Eliminated Razer Synapse dependency by reverse-engineering the local Chroma SDK REST API, implementing BGR encoding and a terminal-driven playlist system with full runtime control over effect order, speed, and direction.

GitHub: github.com/PopcornKitsilano/razer-crimson-royale

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, Java, SQL, Bash

AI and ML: Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, LLM frameworks, LangChain, Prompt Engineering, Model development and deployment, AWS SageMaker, AWS Bedrock, Conversational AI, Generative AI, Supervised and Unsupervised Learning, Feature Engineering

ML Libraries: NumPy, pandas, scikit-learn, Matplotlib, TensorFlow, PyTorch, Hugging Face Transformers

Web and Frameworks: Flask, Django, React, Node.js, HTML5, CSS3, Bootstrap, REST APIs

Databases: SQLite, SQL (schema design, indexing, joins, CRUD operations)

Cybersecurity: Penetration testing, OWASP Top 10, vulnerability assessment, SIEM, log analysis

Security Tools: BurpSuite, Metasploit, Wireshark, Hydra, CAPA, Splunk, Graylog

Developer Tools: Git, GitHub, Linux (Arch, Kali), DaVinci Resolve, Arduino, Agile, VS Code

Concepts: OOP, Data Structures and Algorithms, Software Design, Secure Coding

CERTIFICATIONS

CS50X — Introduction to Computer Science Harvard University via edX View Certificate Verified PDF	Aug 2025
Getting Started with Artificial Intelligence IBM via Credly View Certificate	Nov 2024
PreSecurity Certificate TryHackMe — 100+ rooms; real-world attack simulation, privilege escalation View Certificate	July 2025

HOBBIES

Basketball | [Making YouTube Videos](#) | Music Sampling